

Analysis of a Falling Rigid Rod——Falling Chimneys

Lisu Wang

Catalog

I. Introduction, problem, purpose	3
II. Research & Pre-experiments	3
III. Methods and procedures	7
IV. Result and conclusion	16
V. Future work	17
References	18

I. INTRODUCTION, PROBLEM, PURPOSE

The "falling chimney" is a classic phenomenon in rigid body mechanics. When a tall and thin rod is falling after its base is undermined, it often snaps in the mid-air. This phenomenon is worthy of attention for a safe controlled demolition, which can reduce the risk of damaging surrounding areas, harming people's safety and the unexpected economic losses. So by knowing more about the causes and details of the breaking, we may be able to prevent potential safety consequences.

The purpose of this report is to analyze the internal forces that act on a falling rigid rod to determine why and where the fracture most likely occurs. My hypothesis is that the rod will break at its lower part, and the reason it bends is the internal unbalanced torque due to gravity on different parts.

Furthermore, this report perfectly embodies the three pillars of scientific inquiry: theoretical mathematical derivation, quantitative empirical experimentation using a physical model, and computational verification via a VPython Monte Carlo simulation. By incorporating stochastic material non-uniformity into a computational physics framework, this study bridges abstract theoretical mechanics with contemporary statistical physics and engineering practices.

II. RESEARCH & PRE-EXPERIMENTS

This phenomenon has been studied long in academia. The internal force in the cross-section of the rod is important, including the shear force acting parallel to the plane of the cross-section (perpendicular to the longitudinal axis of the rod), making the two adjacent parts of the material move against each other in opposite directions (Wikipedia contributors, 2024).

Kirk T. McDonald explained that the chimney will break because of these possible reasons:

First, the tension F along the chimney is too great for the bricks to stay attached, but " F is compressive in the case of the falling chimney", so most of the time, we do not attribute the fall to this force.

Second, the shear force, which acts across the slice is too great. Third, the torque τ is too great (2024).

The analysis in the Harvard Department of Physics treats the rod differently. Most models assume the chimney as a thin rod, but this analysis accounts for the chimney's width.

This approach treats the structure as two distinct sub-chimneys connected by longitudinal rods at the edges, allowing for a study of the specific forces. The study notes that at this precise location—one-third of the way up the chimney (from the bottom)—the shear force becomes exactly zero (Morin, 2003).

Another previous work suggests that the portion of the chimney that experiences the biggest stress depends on the angle. At lower angles, the chimney is more likely to break near the middle; at higher angles, the chimney will break at about one third of its height. This phenomenon is known as the Varieschi Drift. (Varieschi et al., n.d.).

With all these previous works, we know that the breaking of the structure in the mid-air is predictable and is related to the geometric distribution of internal stresses. As a result, this paper will explore the fracture height and test the accuracy of these calculations based on AP Physics C: Mechanics, specifically utilizing rotational dynamics and Newton's Second Law for rotation.

What is innovative in this research is the transition from a simplified theoretical model to a Stochastic Finite-Width Model. Unlike previous studies that assume ideal, uniform materials, I designed a simulation that introduces a 10% random variance to the material's structural threshold. This allows for an investigation into the robustness of the model under real-world conditions.

A. Preliminary Qualitative Experiment

To qualitatively observe the phenomenon, a preliminary experiment can be constructed using 36 modular 2×2 LEGO blocks. By building up a vertical rod, this simulates a chimney where the LEGOs' interlocking power represents the tensile strength of mortar.

When tilted and released, while also maintaining the pivot at rest to prevent the sliding of the pivot, the tower consistently experiences mid-air fracture before impact.

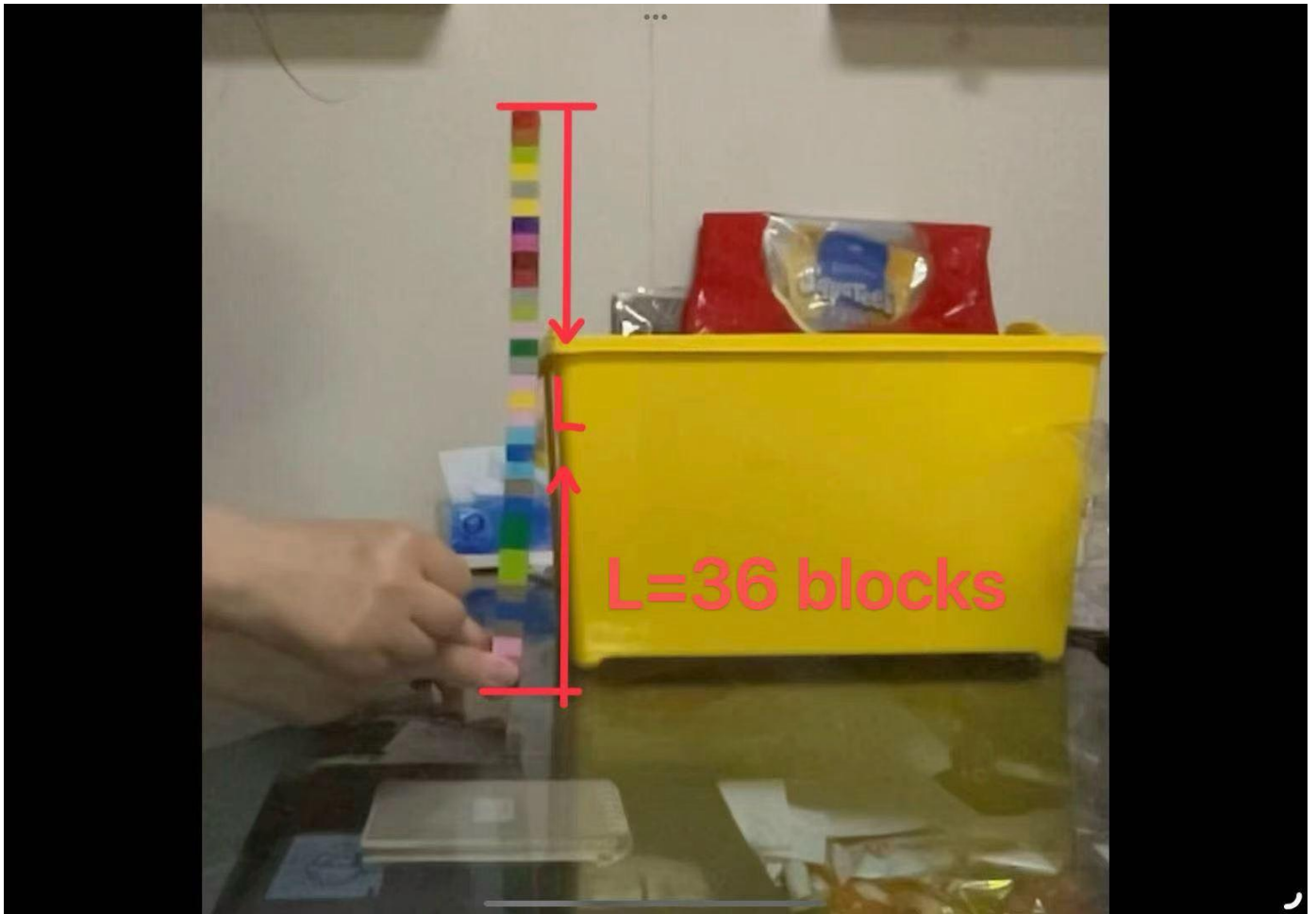


Figure 1. Photograph of the 36-layer LEGO tower before falling

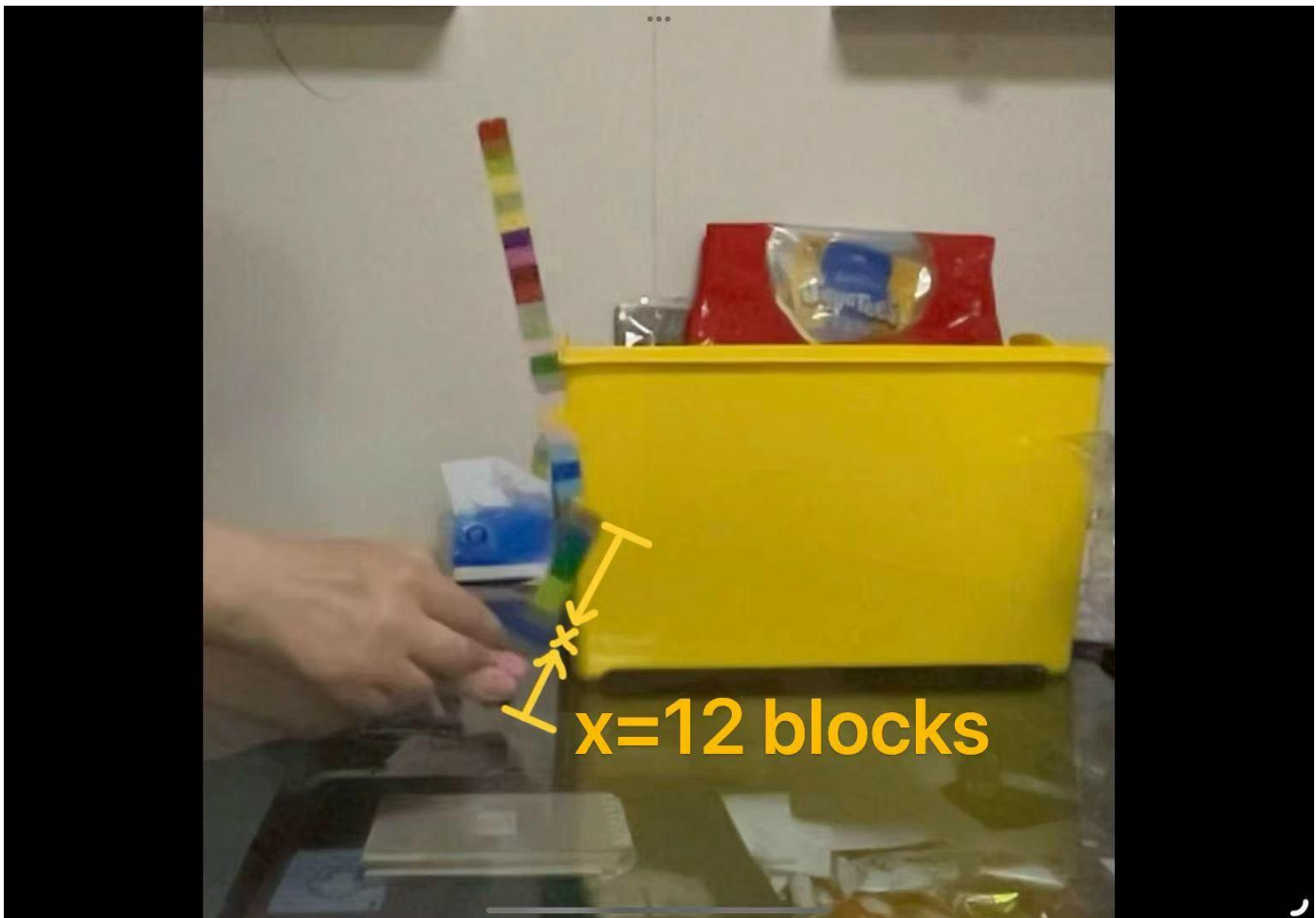


Figure 2. Photograph of the instant that the 36-layer LEGO tower snapped in mid-air

The first separation of blocks occurs approximately 11-12 layers from the base ($x \approx 0.30l - 0.33l$) after several trials (five times). From the observation, we can see the fracture point of the tower is at the lower half of the tower and approximately $\frac{1}{3}$ of the whole length.

To analyze qualitatively, this preliminary experiment demonstrates that the chimney does not break due to the impact with the ground, but rather due to dynamic stresses developed mid-air. The interlocking friction between the LEGO blocks effectively simulates the limited tensile strength of mortar in real masonry chimneys.

As seen in Figure 2, the blocks at the x height mark do not slide past each other (which would indicate shear), but instead pull apart at the leading edge. This provides the qualitative intuition that the rod is experiencing longitudinal tension.

III. METHODS AND PROCEDURES

To simplify the problem, the chimney can be seen as a uniform rod with mass M and length L . Define θ as the angle from the vertical direction. The radius r is small compared to L ($r \ll L$) to describe the overall rotational kinematics, but a finite-width model ($2r$) must be introduced to analyze the internal bending torque into actual tension T_2 at the leading edge later.

First, let's analyze why the rod will break in the mid-air.

A. Rotational Dynamics

The rod rotates about a fixed pivot at its base. The torque τ (around the pivot point) is generated by gravity acting at the center of mass ($\frac{L}{2}$). According to the rotational form of Newton's Second Law $\tau = I\alpha$, and using the moment of inertia for a thin rod ($I = \frac{1}{3}ML^2$):

$$Mg\left(\frac{L}{2}\right) \sin \theta = \left(\frac{1}{3}ML^2\right)\ddot{\theta}$$

Solve the equation for the angular acceleration $\ddot{\theta}$:

$$\ddot{\theta} = \frac{3g \sin \theta}{2L}$$

This equation is important since it can give the tangential acceleration $a_t = x\ddot{\theta}$ at any point x along the chimney. At the top of the chimney ($x=L$), the acceleration is $a_t = \frac{3}{2}g\sin\theta$.

Gravity alone can only provide a maximum downward acceleration of $1g$. For the top part of the chimney to maintain its rigid shape while accelerating at $1.5g$, the lower part of the chimney must actively pull the top part down to force it to follow the rigid body rotation. This internal pulling creates a massive bending moment within the structure. If the material is weak, it cannot sustain the stress required to pull the top end down at $1.5g$. Therefore, the rod will break.

Then, we will find out where the rod breaks.

B. internal stresses

We model the chimney as two segments: a lower portion of height x and an upper portion of length $L-x$ and define the normalized height ratio as $f = \frac{x}{L}$ (where $0 \leq f \leq 1$).

Thus, for the upper segment:

$$length = L - x = L(1-f)$$

$$Mass = M_{upper} = M(1-f)$$

$$Position \text{ for center of mass} = R_{cm} = \frac{x+L}{2} = \frac{L(1+f)}{2}$$

1) Shear force $F_{\perp}$

Using Newton's Second Law in the tangential direction for the upper segment ($F_{net, \perp} = ma_{cm, \perp}$) the forces acting are the internal shear force $F_{\perp}$ and the tangential component of gravity:

$$F_{\perp} + M_{upper}g \sin \theta = M_{upper}R_{cm}\ddot{\theta}$$

Substituting $M_{upper} = M(1-f)$, $R_{cm} = \frac{L(1+f)}{2}$, and $\ddot{\theta} = \frac{3g \sin \theta}{2L}$:

$$F_{\perp} + M(1-f)g \sin \theta = M(1-f) \left[\frac{L(1+f)}{2} \right] \left[\frac{3g \sin \theta}{2L} \right]$$

$$F_{\perp} = \frac{3}{4}Mg(1-f)(1+f) \sin \theta - Mg(1-f) \sin \theta$$

factoring $Mg(1-f) \sin \theta$:

$$F_{\perp}(f) = \frac{Mg \sin \theta}{4} (1-f)[3(1+f)-4]$$

$$F_{\perp}(f) = \frac{Mg \sin \theta}{4} (1-f)(3f-1)$$

2) Bending torque (τ)

Also known as the bending moment, this is the net torque acting across a slice at height x . It exists because the rod's segments have different tangential accelerations.

To find the bending torque, we apply the rotational form of Newton's Second Law $\sum \tau = I_{cm} \alpha$ about the Center of Mass of the upper segment:

$$\tau - F_{\perp} \left(\frac{L-x}{2} \right) = I_{cm, upper} \ddot{\theta}$$

$$\text{where } I_{cm, upper} = \frac{1}{12} M_{upper} (L-x)^2 = \frac{1}{12} M (1-f) [L(1-f)]^2$$

substituting $F_{\perp}$ and simplifying:

$$\tau(f) = \frac{MgL \sin \theta}{4} f(1-f)^2$$

3) Leading-Edge Longitudinal Tension T_2

In the above thin-rod approximation, the fracture is attributed to the internal bending moment (torque effect τ). However, to simulate the situation to a closer detail and to calculate the fracture position at last, the finite width $2r$ of the rod should be considered. The internal torque is resolved into two longitudinal forces acting on the edge—the compression T_1 on the trailing edge, far from ground, and the tension T_2 on the leading edge, close to the ground. As a result, instead of analyzing the torque, we should focus on the tension on the edge when looking closely.

The total torque acting across section is expressed as:

$$\tau = T_1 r + T_2 r$$

It is worth noting that while the compressive force (T_1) and the tensile force (T_2) act in opposite directions along the longitudinal axis, their torques about the neutral axis of the cross-section are in the same direction.

While the sum of $T_1 + T_2$ is in the order of $\frac{1}{r}$, the difference $T_2 - T_1$ is only of order 1, according to the Harvard Model (Morin, 2003). Under the condition that the radius r is much smaller than L ($r \ll L$), the sum of the two forces is large, and difference is small. Mathematically, the two forces are approximately equal ($T_1 \approx T_2$).

Therefore, we are able to get T_2 :

$$T_2(f) = \frac{\tau(f)}{2r} \approx \frac{MgL \sin \theta}{8r} f(1-f)^2$$

Here, a physical clarification must be made to resolve a seeming contradiction with McDonald's model (McDonald, 2024).

McDonald established that when evaluating the chimney as an idealized one-dimensional thin rod, the net longitudinal force (F) integrated across the entire cross-section is compressive, meaning the average axial effect across the slice cannot directly pull the bricks apart (2024).

However, by transitioning to our finite-width model, the macroscopic bending moment (τ) can be resolved into two localized longitudinal boundary forces: a compression T_1 and a tension T_2 .

The true net longitudinal force experienced specifically at the outermost leading edge (T_{net}) is a superposition of this localized tension (T_2) and the global compressive force (F) identified by McDonald (2024). Since force F is driven only by macroscopic gravity and rotation, it remains independent from the localized bending. Because the localized forces T_1 and T_1 add up to zero due to their opposite directions and approximately same magnitude, they do not contribute to McDonald's macroscopic total. Therefore, these two distinct layers of stress—one indicating the overall baseline stress along the rod and the other showing the detailed breakdown at a localized level—are independent and mutually non-inclusive, justifying their mathematical superposition at localized level.

Assuming this total cross-sectional compressive load F is evenly distributed across the two physical boundaries of our finite-width model, the leading edge experiences half of this axial compression. Therefore, the net longitudinal force at the outermost leading edge becomes:

$$T_{\text{net}} = T_2 - \frac{F}{2} = \frac{\tau}{2r} - \frac{F}{2}$$

Under the slender-rod condition ($r \ll L$), the radius r in the denominator amplifies the local bending tension T_2 into a large magnitude, and it eclipses the small distributed compressive contribution of F (which remains within the order of Mg), ensuring that the leading edge remains under a dominant, positive net tension stress ($T_{\text{net}} > 0$).

Since materials like brick or LEGO blocks are weak under tension, this longitudinal tension acting on the outer boundary is the true trigger for the mid-air fracture.

C. Theoretical Prediction and Calculus Validation

While shear force $F_{\perp}$ pushes the bricks together, T_2 attempts to break the joints. Furthermore, in a tall, thin chimney, the magnitude of T_2 is larger than $F_{\perp}$ (Morin, 2003).

Therefore, with the justification of the not compressing nature for longitudinal and eliminating the effect of shear force, to find the point where the rod is most likely to break mid-air, we should find the height ratio f that maximizes T_2 .

In the derived formula for T_2 , the parameters M , g , L , and r are physical constants. While the angle θ from the vertical changes throughout the falling process, it is also treated as a mathematical constant for this derivation because we are analyzing the instant of structural fracture. While θ shows when the structure breaks, the factor $f(1-f)^2$ independently determines where the peak stress occurs.

Expand the factor:

$$G(f) = f(1-f)^2 = f - 2f^2 + f^3$$

Use the first derivative test:

$$\frac{dG}{df} = 1 - 4f + 3f^2$$

To find the extremum,

$$1 - 4f + 3f^2 = 0 \implies (3f-1)(f-1) = 0$$

The solutions are $f = 1$ (at the top of the rod) and $f = \frac{1}{3}$.

This derivation shows that the internal longitudinal tension reaches its theoretical maximum at $\frac{1}{3}$ of the chimney's height. This provides a quantitative explanation for the fracture point.

A significant insight is revealed by substituting the fracture point $f = \frac{1}{3}$ back into the shear force equation $F_{\perp}(f) = \frac{Mg \sin \theta}{4} (1-f)(3f-1)$. The shear force becomes zero. This proves the chimney is not cut apart by shear forces but is pulled apart by tension, which will be the so-called torque effect when ignoring the width of the rod.

However, to account for the Varieschi Drift (Varieschi et al., n.d.), in a real dynamic fall, the net leading-edge stress (T_{net}) is influenced by the axial compression (F) identified by McDonald, which changes during the fall as a function of both $\cos \theta$ and the squaring of angular velocity ω^2 (McDonald, 2024). So the fracture position will be slightly differed from what we derived solely relied on T_2 .

D. Simulations & Experiments

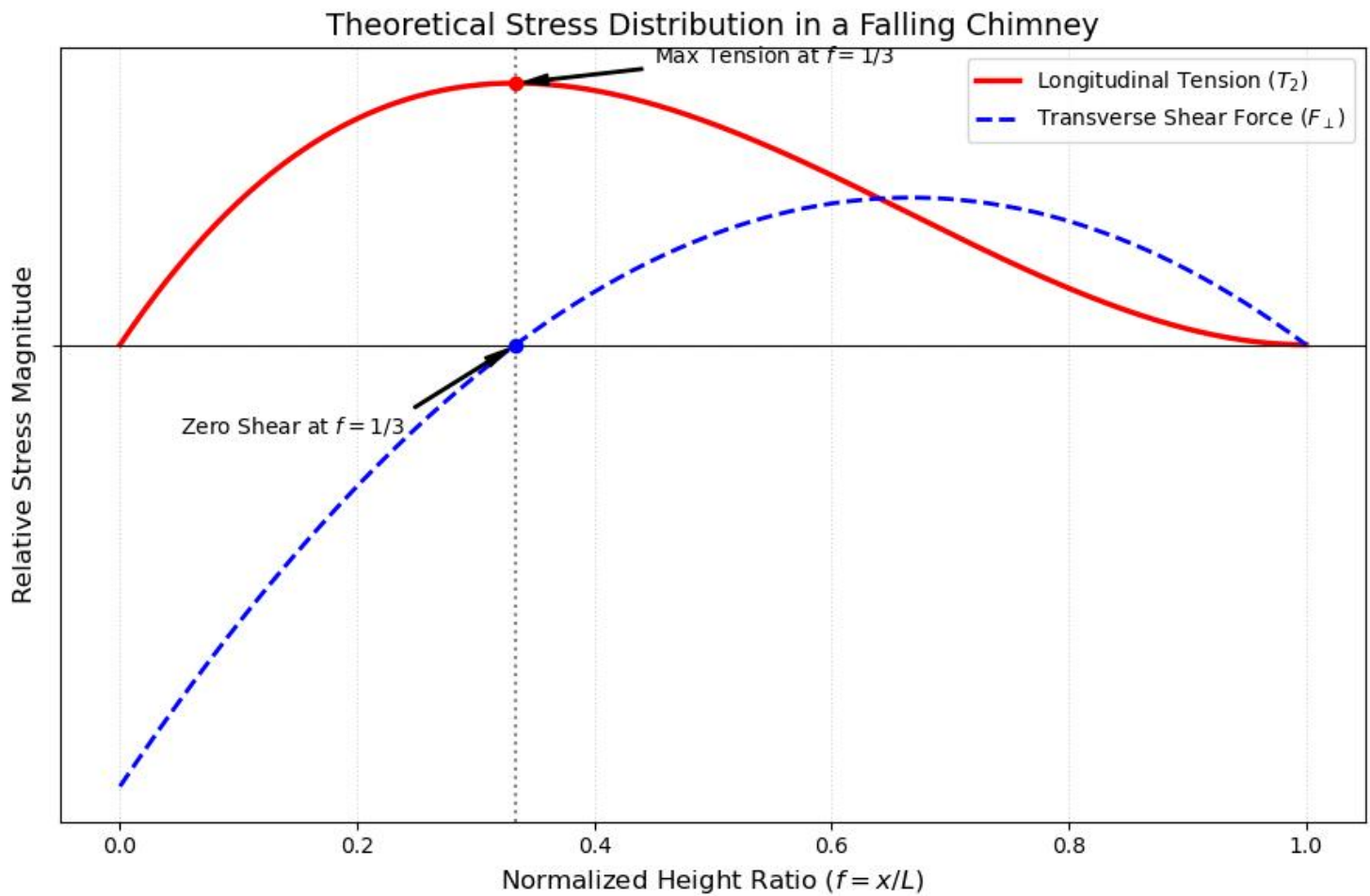


Figure 3. Theoretical Internal Stress Distribution in a Falling Chimney

This graph illustrates the normalized longitudinal tension (T_2) and transverse shear force ($F_{\perp}$) as functions of the height ratio (f). Quantitatively, the longitudinal tension T_2 increases at a decreasing rate from the base and reaches its peak magnitude at $f=0.33$. Simultaneously, the transverse shear force $F_{\perp}$ crosses the zero magnitude threshold. This mathematical alignment confirms that at the point of maximum stress, the shearing effect is negligible, supporting the hypothesis that tension is the primary driver of the failure.

We will use a simulation as the experimentation to test the model.

To verify the model by accounting for real-world material inconsistencies (such as the varying clutch power of bricks), a Monte Carlo simulation was implemented using VPython. A Monte Carlo method is a stochastic computational algorithm that relies on repeated random sampling to obtain numerical results,

particularly useful for systems with significant probabilistic variables. The primary goal was to determine if the $\frac{1}{3}L$ fracture point remains the statistical mode when the material strength is non-uniform.

1) *Description for the simulation*

a) *Discretization of the Rigid Rod*

In each simulation, the chimney of length L is discretized into 36 discrete segments, mirroring the 36-layer LEGO tower used in the pre-experiment. The joints between these segments are the potential points of failure, where the internal stresses are calculated at each time step.

b) *Stochastic Modeling of Tensile Strength (T_{limit})*

In an ideal model, every joint has an identical breaking threshold. However, to account for variations in LEGO bricks or mortar inconsistencies, we defined the Breaking Threshold (T_{limit}) as a random variable following a Normal Distribution, where the mean (μ) is set to a baseline value derived from the maximum calculated T_2 during a successful fall, and the standard deviation (σ) is set to 10% of the mean. This ensures that some joints are slightly stronger or weaker than others.

One might query whether setting the mean (μ) of the Breaking Threshold based on the maximum calculated T_2 introduces circular reasoning. However, the primary objective of this Monte Carlo simulation is to evaluate the robustness of the model against realistic material uncertainties.

We create a hostile environment where non-peak regions could potentially fracture early if they happen to be exceptionally weak.

c) *Algorithmic Loop*

The simulation operates on a nested loop structure.

The Outer Loop accounts for the rotational state of the rod (θ, ω, α) is updated using the differential equation $\ddot{\theta} = \frac{3g \sin \theta}{2L}$ with a time step of $\Delta t = 0.01s$.

The Inner Loop will iterate through all 36 joints ($f = \frac{1}{36}$ to $\frac{35}{36}$) at each time step. For each joint, it calculates the instantaneous Leading-Edge Longitudinal Tension (T_2): $T_2(f, \theta) = \frac{MgL \sin \theta}{8r} f(1-f)^2$

If $T_2(f, \theta) > T_{\text{limit, joint}}$, the simulation immediately triggers a snap event. The normalized height (f) and the angle of fracture (θ_{break}) are recorded, and the current trial is terminated.

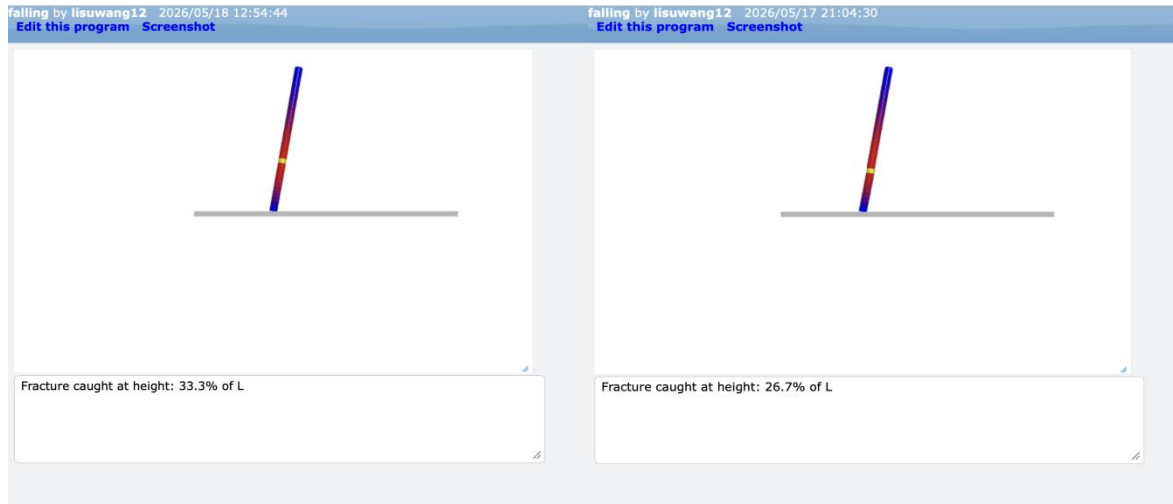


Figure 4. Demonstration of two trials in the Monte Carlo simulation showing fracture caught at 33.3%L and 26.7%L, respectively, due to a localized material weakness.

2) Iterative Validation (100 Trials)

The Monte Carlo method involves running this sequence 100 independent times to form a probability distribution.

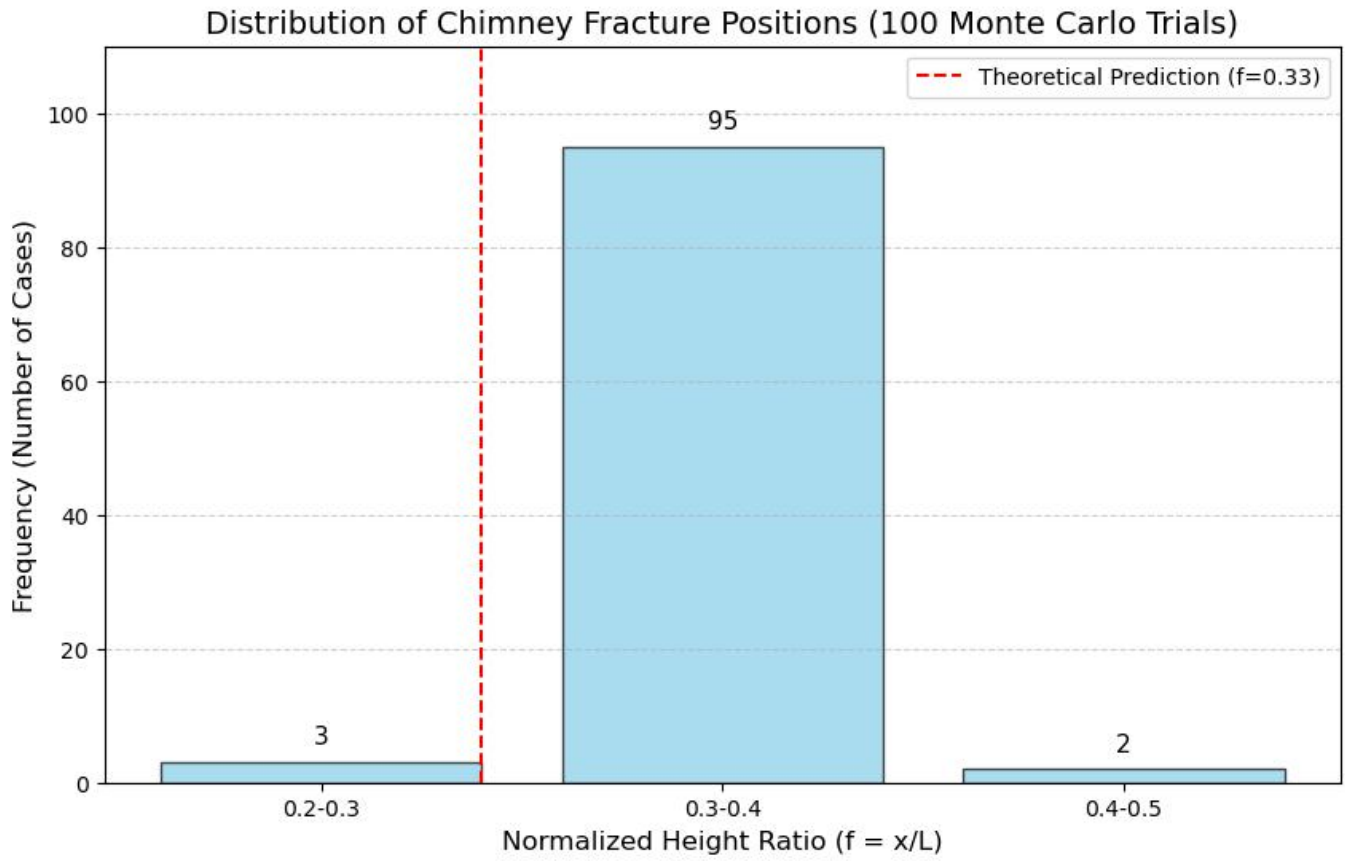


Figure 5. The histogram displays the frequency of fractures.

With 95% of cases occurring in the $0.3L$ – $0.4L$ range despite the 10% material randomness, it provides overwhelming evidence to validate the model, demonstrating that the geometric distribution of the bending moment is the dominant factor in structural fracture, rather than localized material inconsistencies.

IV. RESULT AND CONCLUSION

Method	Methodology Type	Expected / Average Fracture Position (f)
Calculus Analytical Model	Theoretical Derivation	0.333 ($\frac{1}{3} L$)
VPython Monte Carlo Simulation	Computational Simulation (100 Trials)	0.3 - 0.4 (Statistical Center)
LEGO Tower Drop	Preliminary Physical Experiment (5 Trials)	0.311 (Layers 11–12)

Table 1. Quantitative Comparison of Fracture Location

As summarized in Table 1, a comprehensive cross-Method analysis shows consistency on the fracture position at $f=0.333$.

To test the robustness of this model against real-world material uncertainties, the 100-trial VPython Monte Carlo simulation (illustrated in Figure 5) was analyzed. The results yield a normal distribution centered at $f=0.3$ to 0.4 , proving that localized structural weaknesses do not shift the macro-level peak of the bending moment.

Also the previous physical trials using the 36-layer LEGO tower provide direct macroscopic validation, with an average snapping point observed at $f \approx 0.311$. The minor discrepancy ($\approx 6.6\%$) between the physical trials and the theoretical baseline can be justified by two factors: air resistance and the discrete interlocking joints of the LEGO bricks which create structural stress concentration points.

The hypothesis in section I stated that the rod would break at its lower part. Since $f < 0.5$ denotes the bottom half of the structure, the concentration of structural fracture inside the lower one-third region proves the positional hypothesis to be accurate.

Also, the finding delves deeper into the cause predicted in the hypothesis, which argues that the fracture is brought by internal unbalanced torque due to gravity on different parts. This macroscopic bending torque is analyzed as the longitudinal tension that pulls the structure apart.

V. FUTURE WORK

To enhance the ability and accuracy of the current model, future research could incorporate non-uniform mass distributions, for example rods where the linear mass density $\lambda(x)$ changes with height. Also, accounting for friction at the pivot point could also bring potential new insights. Finally, investigating the fracture height under Varieschi Drift by testing various initial release angles can also improve the performance for the model.

REFERENCES

- [1] Wikipedia contributors. (2024, May 10). Shear force. In Wikipedia, The Free Encyclopedia.
https://en.wikipedia.org/wiki/Shear_force
- [2] McDonald, K. T. (2024, November 21). Falling chimney. Joseph Henry Laboratories, Princeton University.
<https://kirkmc.princeton.edu/examples/chimney.pdf>
- [3] Week 49: Falling chimney [Solution]. (2003, August 18). Harvard University Department of Physics.
<https://www.physics.harvard.edu/sites/g/files/omnuum6476/files/physics/files/sol49.pdf>
- [4] Varieschi, G., Kamiya, K., & Jully, I. (n.d.). The falling chimney web page. Department of Physics, Loyola Marymount University. <http://webpages.lmu.edu/gvarieschi/chimney/chimney.html>